

# Foundations of Time series forecasting

[Time series forecasting]

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## 1.0 Content Block

Motivation In the context of statistics, econometrics, quantitative finance, seismology, meteorology, and geophysics the primary goal of time series analysis is forecasting. In the context of signal processing, control engineering and communication engineering it is used for signal detection. Other applications are in data mining, pattern recognition and machine learning, where time series analysis can be used for clustering, classification, query by content, anomaly detection as well as forecasting.

## 2.0 Content Block

A simple way to examine a regular time series is manually with a line chart. The datagraphic shows tuberculosis deaths in the United States, along with the yearly change and the percentage change from year to year. The total number of deaths declined in every year until the mid-1980s, after which there were occasional increases, often proportionately - but not absolutely - quite large. A study of corporate data analysts found two challenges to exploratory time series analysis: discovering the shape of interesting patterns, and finding an explanation for these patterns. Visual tools that represent time series data as heat map matrices can help overcome these challenges.

## 3.0 Content Block

For processes that are expected to generally grow in magnitude one of the curves in the graphic (and many others) can be fitted by estimating their parameters. The construction of economic time series involves the estimation of some components for some dates by interpolation between values ("benchmarks") for earlier and later dates. Interpolation is estimation of an unknown quantity between two known quantities (historical data), or drawing conclusions about missing information from the available information ("reading between the lines"). Interpolation is useful where the data surrounding the missing data is available and its trend, seasonality, and longer-term cycles are known. This is often done by using a related series known for all relevant dates. Alternatively polynomial interpolation or spline interpolation is used where piecewise polynomial functions are fitted in time intervals such that they fit smoothly together. A different problem which is closely related to interpolation is the approximation of a complicated function by a simple function (also called regression). The main difference between regression and interpolation is that polynomial regression gives a single polynomial that models the entire data set. Spline interpolation, however, yield a piecewise continuous function composed of many polynomials to model the data set. Extrapolation is the process of estimating, beyond the original observation range, the value of a variable on the basis of its relationship with another variable. It is similar to interpolation, which produces estimates between known observations, but extrapolation is subject to greater uncertainty and a higher risk of producing meaningless results.

## 4.0 Content Block

Univariate linear measures Moment (mathematics) Spectral band power Spectral edge frequency Accumulated energy (signal processing) Characteristics of the autocorrelation function Hjorth parameters FFT parameters Autoregressive model parameters Mann–Kendall test Univariate non-linear measures Measures based on the correlation sum Correlation dimension Correlation integral Correlation density Correlation entropy Approximate entropy Sample entropy Fourier entropy Wavelet entropy Dispersion entropy Fluctuation dispersion entropy Rényi entropy Higher-order methods Marginal predictability Dynamical similarity index State space dissimilarity measures Lyapunov exponent Permutation methods Local flow Other univariate measures Algorithmic complexity Kolmogorov complexity estimates Hidden

Markov model states Rough path signature Surrogate time series and surrogate correction Loss of recurrence (degree of non-stationarity) Bivariate linear measures Maximum linear cross-correlation Linear Coherence (signal processing) Bivariate non-linear measures Non-linear interdependence Dynamical Entrainment (physics) Measures for phase synchronization Measures for phase locking Similarity measures: Cross-correlation Dynamic time warping Hidden Markov model Edit distance Total correlation Newey–West estimator Prais–Winsten transformation Data as vectors in a metrizable space Minkowski distance Mahalanobis distance Data as time series with envelopes Global standard deviation Local standard deviation Windowed standard deviation Data interpreted as stochastic series Pearson product-moment correlation coefficient Spearman's rank correlation coefficient Data interpreted as a probability distribution function Kolmogorov–Smirnov test Cramér–von Mises criterion

## 5.0 Content Block

In mathematics, a time series is a sequence of data points indexed, listed, or graphed in chronological order. Most commonly, a time series consists of observations recorded at successive equally spaced points in time. Thus, it represents a form of discrete-time data. A time series may describe measurements collected over seconds, days, years, or even centuries. Common examples include heights of ocean tides, counts of sunspots, daily temperature readings, and the closing values of stock market indices such as the Dow Jones Industrial Average. A time series is often visualized using a run chart (a type of temporal line chart), which helps identify patterns such as trends, seasonal effects, and irregular fluctuations. Time series are widely used in statistics, actuarial science, signal processing, pattern recognition, econometrics, mathematical finance, weather forecasting, earthquake prediction, electroencephalography, control engineering, astronomy, communications engineering, and many other areas of applied science and engineering that involve temporal measurements. Time series analysis comprises methods for analyzing time series data in order to extract meaningful statistics and other characteristics of the data. Time series forecasting is the use of a model to predict future values based on previously observed values. Generally, time series data is modeled as a stochastic process. While regression analysis is often employed in such a way as to test relationships between one or more different time series, this type of analysis is not usually called "time series analysis", which refers in particular to relationships between different points in time within a single series. Time series data have a natural temporal ordering. This makes time series analysis distinct from cross-sectional studies, in which there is no natural ordering of the observations (e.g. explaining people's wages by reference to their respective education levels, where the individuals' data could be entered in any order). Time series analysis is also distinct from spatial data analysis where the observations typically relate to geographical locations (e.g. accounting for house prices by the location as well as the intrinsic characteristics of the houses). A stochastic model for a time series will generally reflect the fact that observations close together in time will be more closely related than observations further apart. In addition, time series models will often make use of the natural one-way ordering of time so that values for a given period will be expressed as deriving in some way from past values, rather than from future values (see time reversibility). Time series analysis can be applied to real-valued, continuous data, discrete numeric data, or discrete symbolic data (i.e. sequences of characters, such as letters and words in the English language).